

# CAREER SUMMARY

## **Olufemi Osaloni, Ph.D. (ABD) | Senior Data Scientist & AI/ML Engineer**

My career journey represents a strategic evolution from traditional electrical engineering to the cutting edge of artificial intelligence and data science. What began as a passion for solving complex engineering challenges has transformed into a mission to leverage AI and machine learning to drive measurable business impact across industries.

### **Engineering Foundation & Industry Impact**

I launched my career at Eaton Corporation in Milwaukee, WI, where I led the design, testing, and qualification of grid-tie solar inverters up to 500kW capacity. This role demanded not only technical expertise in power systems and renewable energy but also the ability to navigate complex industry standards (UL1741, IEEE1547) and deliver commercial-scale solutions. The experience instilled in me a deep appreciation for precision engineering and the importance of rigorous testing methodologies.

Driven by my love for taking on challenges, I made a strategic move to FMC Technologies' subsea division—a specialized branch of their innovation and R&D business unit. The mission was ambitious: design a subsea inverter to power Christmas trees at the bottom of the ocean. This role pushed the boundaries of what was possible in harsh environments and required innovative thinking in power systems design. During this time, I developed conceptual engineering designs using advanced simulation tools and MATLAB/Simulink, and led initiatives to identify disruptive technologies in the subsea power domain.

### **The Pivot: Discovering Data Science**

While working on subsea power systems, I became captivated by data science—a disruptive technology that was reshaping industries. I recognized that the analytical mindset I had developed as an engineer could be amplified through machine learning and advanced analytics. This revelation led me to pursue a Master's in Analytics from Villanova University (graduating with a perfect 4.0 GPA) and transition fully into data science and machine learning engineering.

### **Delivering Value Through Data Science & Machine Learning**

Since embracing data science, I have worked across multiple organizations—TechnipFMC and LyondellBasell Industries—where I have consistently delivered quantifiable business value:

- Achieved over \$7M in cost savings through predictive maintenance, inventory optimization, and intelligent automation projects
- Developed end-to-end ML recommendation systems leveraging Sentence-BERT embeddings, reducing excess inventory and saving \$5M+
- Built enterprise-scale analytics platforms on AWS (SageMaker, Lambda, Glue) and Databricks, improving data processing speed by 50% and enabling real-time monitoring across global operations
- Reduced unplanned downtime by 25% through statistical modeling, anomaly detection, and predictive analytics, delivering \$2M+ in operational savings
- Led cross-functional initiatives translating advanced ML solutions into measurable business impact, mirroring a consulting engagement model

### **Current Focus: Generative AI & Large Language Models**

As a work hobby and professional passion, I have immersed myself in the world of Generative AI and Large Language Models (LLMs)—the next frontier of AI innovation. I am actively exploring and working with cutting-edge frameworks including:

- LangChain, LangSmith, and LangGraph for building sophisticated agentic workflows
- OpenAI Agents SDK for intelligent automation and autonomous systems
- Model Context Protocol (MCP) for seamless model integration and orchestration
- Crew AI and AutoGen for multi-agent collaboration and complex task automation
- Retrieval-Augmented Generation (RAG) systems with vector databases (FAISS, ChromaDB) for enhanced knowledge retrieval

I have designed and implemented RAG prototypes with Streamlit interfaces to automate document search and answer generation, improving retrieval speed and accuracy for internal workflows. My goal is to grow expertise in this transformative field and successfully apply Gen AI solutions across diverse industries—just as I have done throughout my career as a data scientist using AWS, Azure, and other enterprise platforms.

## Innovation & Intellectual Property

My commitment to innovation is reflected in my patent for a **Maintenance Condition Sensing Device (US20180095455A1)**, which demonstrates my ability to identify operational challenges and develop novel technical solutions that advance industry standards.

## Vision for the Future

My career has been defined by a willingness to embrace new challenges and a relentless focus on delivering results. From designing solar inverters and subsea power systems to building enterprise ML platforms and exploring Gen AI frontiers, I have consistently pushed the boundaries of what technology can achieve. As I continue to deepen my expertise in LLMs and agentic AI frameworks, I am excited to apply these capabilities to solve complex problems across industries, driving innovation and creating lasting business value.